

# James Kurtis

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## EDUCATION

**Cornell University**, College of Engineering, Ithaca, NY

*Expected May 2028*

*B.S. Mechanical Engineering, Business Minor for Engineers* | GPA: 3.68

Relevant Courses: Differential Equations, Statics & Mechanics of Solids, Dynamics, Thermodynamics, Intro to Mechanical Design, Engineering Simulations & Design, Physics I, II, III, System Dynamics, Fluid Dynamics, Mechanics of Materials.

## TECHNICAL EXPERIENCE

**Cornell Racing FSAE Project Team**, Cornell University, *Cooling Sub-team Lead*

*Oct 2024-Present*

- Lead a 5-person cooling sub-team, hosting weekly design meetings and delegating tasks while coordinating integration with other sub-teams to develop full-car cooling infrastructure for Cornell Racing's electric race car.
- Built a MATLAB model of the car's liquid cooling loop to predict required flow rates and component pressure drops, aiming to minimize system mass by providing part designers with optimized design targets.
- Fitted the previous cooling system with temperature and flow rate sensors to validate the team's design methodology against endurance performance data.
- Redesigned and manufactured the tractive battery container, saving 1kg (10%) of weight from the previous design through iterative modeling with Autodesk Inventor.
- Created GD&T-compliant engineering drawings for battery mounting hardware and machined clevises, bushings, and other aluminum parts on a mill and lathe.
- Performed internal resistance testing on lithium-polymer pouch cells to quantify heat generation, validating battery pack architecture, cooling and BMS configuration.

**Ametek Hughes-Treitler**, Garden City, NY, *Materials Management Intern*

*Jun-Aug 2026*

- Manufactured inconel foil (laser-cutting, stamping, cleaning) and assembled stacked-foil cores for 5 iterations of experimental micro-foil heat exchangers to support performance and pressure testing.
- Reverse-engineered performance specifications (flow rate, pressure drop, heat rejection) from 20+ competitor direct-to-chip GPU cold plates, informing cold plate redesigns and next-gen performance targets.
- Built 6 Power BI/Excel tools to consolidate sales, demand, and inventory data, aligning planning, purchasing, and manufacturing teams on production decisions while saving planners ~3 hours per week.

**Aesthetic Wrks**, Hempstead, NY, *Car Detailing Apprenticeship*

*Jun-Aug 2025*

- Shadowed technicians during automotive maintenance including oil changes, spark plug replacement, and basic diagnostics.
- Installed vinyl wraps and paint protection film and performed minor bodywork on high-performance and luxury vehicles, building hands-on familiarity with diverse vehicle systems.

## LEADERSHIP EXPERIENCE

**Varsity Wrestling**, Xavier High School, *Team Captain*

*Nov 2022-Feb 2024*

- Elected by teammates to lead and instruct a 60-person team, strengthening leadership and communication skills.
- Qualified for the CHSAA State Championship and competed against top wrestlers in the state.

**Valor Mixed Martial Arts**, First Degree Black Belt, NY

*2012 - Present*

- Achieved 1st degree black belt in Kaju Bujutsu Kwai, reflecting 13 years of sustained discipline and perseverance.
- Mentored junior students and performed on the leadership team, supporting team development.

## TECHNICAL SKILLS

**CAD/Simulation:** ANSYS, Autodesk Inventor, Autodesk Fusion 360, Rhino 7

**Manufacturing:** Machining (Mill, Lathe), Laser Welding (Lightweld 2000XR), 3D Printing, Soldering, GD&T Drawings

**Programming & Tools:** Python, MATLAB, Microsoft Office (Word, Excel, PowerPoint, Power BI, Power Automate)

## INTERESTS

Formula One • Electric Vehicles • Mixed Martial Arts • Karting • Scuba Diving (PADI Advanced Open Water) • Guitar